

TCC70xx MCU BSP

API Specification for Input Capture Time Counter

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1 INTRODUCTION

This document describes the input capture time counter device driver (hereinafter referred to as ICTC driver) of TCC70xx MCU BSP. For more detailed information of the input capture time counter in the TCC70xx MCU Sub-system, refer to Chapter 13 Input Capture Time Counter (ICTC) in "*TCC70xx Full Specification*"[1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

ICTC	Input Capture Time Counter
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3 SOURCE FILES

3.1 Location of Source Files

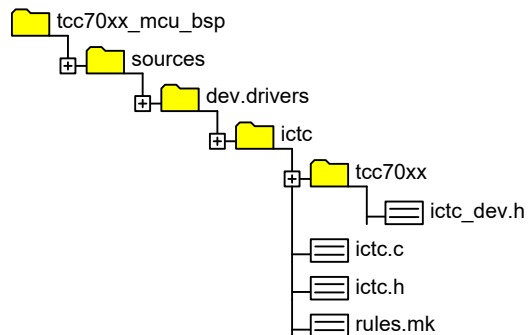


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- `ictc_dev.h`
 - ICTC specific header
- `ictc.c`
 - ICTC common source code
- `ictc.h`
 - ICTC common header including API & Defines
- `rules.mk`
 - Rules used for build

4 CONSTANTS

4.1 ICTC Channel Index

Constant values for the ICTC channel index

Declaration

```
#define ICTC_CH_0 (0UL)
#define ICTC_CH_1 (1UL)
#define ICTC_CH_2 (2UL)
#define ICTC_CH_3 (3UL)
#define ICTC_CH_4 (4UL)
#define ICTC_CH_5 (5UL)
```

Description

Value	Description
ICTC_CH_0	Index of ICTC channel0
ICTC_CH_1	Index of ICTC channel1
ICTC_CH_2	Index of ICTC channel2
ICTC_CH_3	Index of ICTC channel3
ICTC_CH_4	Index of ICTC channel4
ICTC_CH_5	Index of ICTC channel5

Remark

None

4.2 Channel Information Value

Constant values for ICTC channel information value

Declaration

```
#define ICTC_SUB_ICTC_ENABLED
#define ICTC_TOTAL_CHANNEL_NUM      (6UL)
#define ICTC_DIFF_MARGIN            (0x8000UL)

#define ICTC_TIMESTAMP_RECORD_ENABLE
#define ICTC_TIMESTAMP_RECORD_MAXCNT (15UL)
```

Description

Value	Description
<i>ICTC_SUB_ICTC_ENABLED</i>	<i>Enable Redundancy ICTC for safety feature</i>
<i>ICTC_TOTAL_CHANNE_NUM</i>	<i>Total count of the ICTC channels</i>
<i>ICTC_DIFF_MARGIN</i>	<i>Margin Value between Main ICTC and Sub-ICTC for safety feature</i>
<i>ICTC_TIMESTAMP_RECORD_ENABLE</i>	<i>Enable Time-stamp Record</i>
<i>ICTC_TIMESTAMP_RECORD_MAXCNT</i>	<i>Maximum count of Recorded Time-stamps</i>

Remark

None

5 STRUCTURES

5.1 ICTC callBack_t

Structure to register a callback function that notifies you of ICTC interrupt handle completion

Declaration

```
typedef struct _ICTCcallBack_  
{  
    int32          cbChannel;  
    ICTCcallback   cbCallBackFunc;  
} ICTCcallBack_t;
```

Description

Value	Description
uiChannel	ICTC channel Number 0 to 5
ucCallBackFunc	Pointer of callback function. Refer to ICTCcallBack .

Remark

None

6 CALLBACK FUNCTIONS

6.1 ICTCCallback

When the ICTC input interrupt handle is done, the ICTC driver calls this function to deliver the calculated period and duty value to you.

Declaration

```
typedef void (*ICTCCallback)
(
    uint32 uiChannel,
    uint32 uiPeriod,
    uint32 uiDuty
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in] ICTC channel Number 0 to 5</i>
<i>uiPeriod</i>	<i>[in] Period value of input signal. This value is delivered to you.</i>
<i>uiDuty</i>	<i>[in] Duty value of input signal. This value is delivered to you.</i>

Return

None

Remark

None

7 APIs

7.1 ICTC_Init

Enables peripheral clock for the ICTC module.

Declaration

```
void ICTC_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.2 ICTC_Deinit

Disables peripheral clock for the ICTC module.

Declaration

```
void ICTC_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.3 ICTC_EnableCapture

Starts the input signal capture.

Declaration

```
void ICTC_EnableCapture
(
    uint32    uiChannel
);
```

Parameters

Value	Description
uiChannel	[in] ICTC channel Number 0 to 5

Return

None

Remark

None

7.4 ICTC_DisableCapture

Stops the input signal capture.

Declaration

```
void ICTC_DisableCapture
(
    uint32    uiChannel
);
```

Parameters

Value	Description
uiChannel	[in] ICTC channel Number 0 to 5

Return

None

Remark

None

7.5 ICTC_SetOpEnCtrlCounter

Sets the ICTC counter types in *OP_EN_CTRL* register.

Declaration

```
void ICTC_SetOpEnCtrlCounter
(
    uint32    uiChannel,
    uint32    uiCntEnValue
);
```

Parameters

Value	Description
<i>uiChannel</i>	[in] ICTC channel Number 0 to 5
<i>uiCntEnValue</i>	[in] Counter enable value which is defined in <i>ictc_dev.h</i> (<i>ICTC_OPEN_CTRL_XXXX_EN</i>)

Return

None

Remark

There are six counter types.

- Pre-scaler counter (TCLK_EN)
- Time stamp counter (TSCNT_EN)
- Filter counter (FLTCNT_EN)
- Time-out counter (TOCNT_EN)
- Edge counter (ECNT_EN)
- Period/duty compare counter (PDCNT_EN)

7.6 ICTC_SetOpModeCtrlReg

Sets the ICTC mode and input port (*F_IN_SEL*) in *OP_MODE_CTRL* register.

Declaration

```
sint32 ICTC_SetOpModeCtrlReg
(
    uint32 uiChannel,
    uint32 uiModeValue
);
```

Parameters

Value	Description
<i>uiChannel</i>	[in] ICTC channel Number 0 to 5
<i>uiModeValue</i>	[in] Mode value which is defined in <i>ictc_dev.h</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	Success
<i>SAL_RET_FAILED</i>	Failure

Remark

Refer to Chapter 13 in "*TCC70xx Full Specification*" for MODE information.

7.7 ICTC_SetIRQCtrlReg

Enables the ICTC interrupt. The ICTC_SetIRQCtrlReg sets the selected IRQ types in *IRQ_CTRL* register (IRQ_EN[31:24]).

Declaration

```
void ICTC_SetIRQCtrlReg
(
    uint32 uiChannel,
    uint32 uiIRQValue
);
```

Parameters

Value	Description
uiChannel	[in] ICTC channel Number 0 to 5
uiIRQValue	[in] IRQ type value

Return

None

Remark

Refer to Chapter 13 in "*TCC70xx Full Specification*" to get *IRQ_EN* information.

7.8 ICTC_SetCallbackFunc

Registers the ICTC callback function.

Declaration

```
void ICTC_SetCallbackFunc
(
    uint32      uiChannel,
    ICTCCallback pCallbackFunc
);
```

Parameters

Value	Description
uiChannel	[in] ICTC channel Number 0 to 5
pCallbackFunc	[in] The pointer of callback function. Refer to ICTCCallback .

Return

None

Remark

None

7.9 ICTC_SetTimeoutValue

Sets the ICTC timeout value. If input signal is slower than ICTC clock (Example: ICTC: 100 MHz/input: 5 MHz), the timeout count increases. Then, when the timeout count reaches timeout value, a timeout interrupt is generated.

Declaration

```
void ICTC_SetTimeoutValue
(
    uint32 uiChannel,
    uint32 uiTimeoutValue
);
```

Parameters

Value	Description
uiChannel	[in] ICTC channel Number 0 to 5
uiTimeoutValue	[in] ICTC Timeout value

Return

None

Remark

None

7.10 ICTC_SetEdgeMatchingValue

Sets the ICTC edge matching values. At the start of capture, the period is ignored as much as the ICTC edge matching value set. This is to deal with the noise that occurs at the start of the capture.

Declaration

```
void ICTC_SetEdgeMatchingValue
(
    uint32 uiChannel,
    uint32 uiRisingEdgeMat,
    uint32 uiFallingEdgeMat,
    uint32 uiEdgeCounterMat
);
```

Parameters

Value	Description
uiChannel	[in] ICTC Channel value 0 to 5
uiRisingEdgeMat	[in] Rising edge matching value
uiFallingEdgeMat	[in] Falling edge matching value
uiEdgeCounterMat	[in] Edge counter matching value

Return

None

Remark

None

7.11 ICTC_SetCompareRoundValue

Sets the ICTC period and duty compare round values (*PRD_CMP_RND*, *DUTY_CMP_RND*). The *PRD_CMP_RND* means a frequency counter masking value for comparing the previous frequency with the current frequency. If the previous frequency counter value and the current frequency counter value are not the same, an ICTC interrupt occurs. The *DUTY_CMP_RND* is similar to *PRD_CMP_RND*

Declaration

```
void ICTC_SetCompareRoundValue
(
    uint32 uiChannel,
    uint32 uiPeriodCompareRound,
    uint32 uiDutyCompareRound
);
```

Parameters

Value	Description
uiChannel	[in] ICTC Channel value 0 to 5
uiPeriodCompareRound	[in] Period compare round value
uiDutyCompareRound	[in] Duty compare round value

Return

None

Remark

None

8 HOW TO USE ICTC DRIVER

You should initialize and configure the ICTC module before starting input capture. Refer to *ictc_test.c* file to get example code.

8.1 Operation of ICTC

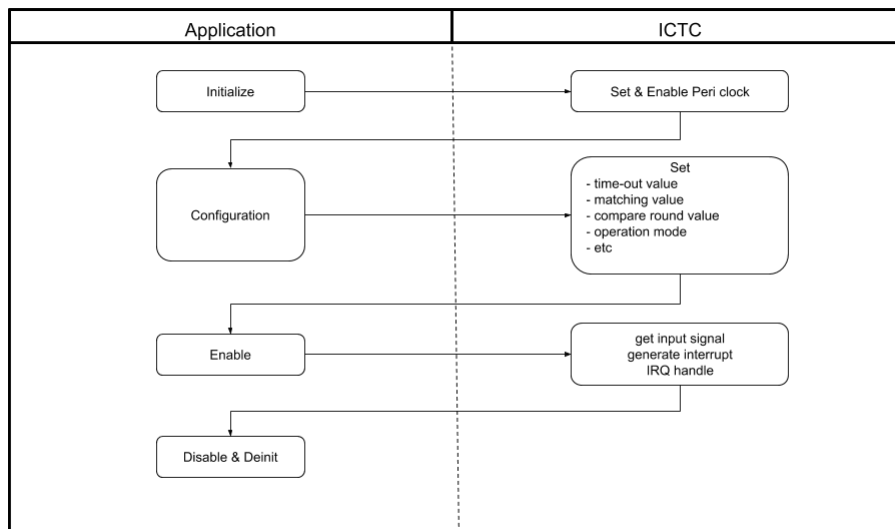


Figure 8.1 Diagram for ICTC Driver

8.2 Example: Using ICTC Driver

```

{
    //Initialize
    ICTC_Init();

    //Configuration
    ICTC_TestSetGpio();

    ICTC_SetCallBackFunc(...);      //optional
    ICTC_SetTimeoutValue(...);      //optional
    ICTC_SetEdgeMatchingValue(...); //optional
    ICTC_SetCompareRoundValue(...); //optional

    ICTC_SetOpModeCtrlReg(...);
    ICTC_SetOpEnCtrlCounter(...);
    ICTC_SetIRQCtrlReg(...);

    // Enable (Start input signal capture)
    ICTC_EnableCapture();

    // ...
    // process IRQ generated from duty & period change of input signal

    // Disable (Stop input signal capture)
    ICTC_DisableCapture();
}
  
```

9 REFERENCES

- [1] TCC70xx Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

10 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release

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